Chittilla Sumedh Srujan

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Msc Tech Physics
National Institute Of Technology Warangal

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Education

•National Institute of Technology Warangal *2027*
Msc Tech Physics CGPA/Percentage: 7.9

•Amrita Vishwa Vidyapeetham Coimbatore *2023*
Bsc Physics CGPA/Percentage: 7.54

Experience

•amPICQ Pvt. Ltd. *February 2026 Present*
Research Intern Photonic Integrated Circuits Remote

- Designed and verified receiver side Silicon Nitride Photonic Integrated Circuits for secure optical communications and Quantum Key Distribution systems, delivering foundry ready GDSII layouts with optimized MZI networks and phase shifters.
- Handled electro optical characterization of receiver chips, managing fiber to chip alignment, phase modulator voltage and current sweeps, and wavelength tuning to extract extinction ratios, $V_{2\pi}$, FSR, and thermo optic coefficients.
- Conducted photon budget and key rate calculations for QKD protocols, optimizing detector timing windows and modulator specifications.
- *Status: Ongoing | Tools: Lumerical INTERCONNECT, FDTD, MODE, GDSII Layout Tools, Optical Bench Characterization*

•Zen Technologies Limited *Ongoing*
Optical Design intern

- Designing a highly integrated long range 360° multi camera array for defense and anti drone applications ,focusing on wide field and high resolution spatial sampling.

•Indian Institute of Science Bangalore (IISc Bangalore) *December 2025 June 2026*
Research Intern Optical Design Bangalore, India

Design of Projection Lens for UV based Maskless Lithography
- Designed a high precision projection lens for DMD based maskless lithography using Zemax OpticStudio.
- Optimized a multi element projection lens system to correct complex optical aberrations, achieving a diffraction limited configuration.
- Ran detailed geometric image simulations and successfully validated spatial resolution limits using both Modulation Transfer Function analysis and image simulation tools.
- *Simulation Tools: Zemax OpticStudio*

Super Oscillatory Metalens Focal Spot Characterization
- Set up a 405 nm optical characterization bench for super oscillatory metalens point spread function measurement, targeting focal spot verification at 200 nm resolution (0.49λ, NA ≈ 1.0).
- Engineered the illumination path using far field Gaussian approximation of a 405 nm laser diode output with two mirror beam walking for on axis illumination across a 2 mm SOL aperture.
- Designed and aligned a 4f relay (high NA objective, tube lens, M $\approx 100\times$) to image the SOL focal plane onto a scientific camera for central hotspot FWHM measurement and side lobe ring characterization.
- Positioned a precision pinhole at the relay conjugate image plane to spatially filter the side lobe ring and quantify central lobe throughput versus background rejection ratio.

Polarization Multiplexed Bifocal Metalens Design & Simulation
- Designed an all dielectric 15 μm metalens comprising 4,421 TiO_2 nanopillars on a SiO_2 substrate to independently focus orthogonal polarizations into distinct spots 6 μm apart at 532 nm.
- Ran full wave electromagnetic simulations using Lumerical FDTD, achieving $\sim$44% focusing efficiency and a diffraction limited spot size of 0.52 μm (within 4% of the fundamental limit).
- Eliminated spurious diffraction orders by synthesizing independent off axis phase profiles rather than conventional grating superposition, balancing polarization channels to <1% via joint dimensional optimization.

UV Projection Lens Design for DLP Micro Stereolithography (μSLA) 3D Printer

- Designed a diffraction limited 2× telecentric UV projection lens at 405 nm using stock Thorlabs catalog optics in Zemax OpticStudio, achieving RMS spot radius $\leq 15.4\,\mu$m across a 17.5 mm object field.
- Constrained all elements to stock Thorlabs UV catalog lenses (fused silica, CaF_2) with no custom optics, demonstrating a cost aware optical design approach.
- Built custom Merit Functions combining PMAG, REAB telecentricity operands, RSCE spot constraints, and boundary guards to simultaneously control magnification, image side telecentricity, and aberration correction.

•**Indian Institute of Technology Delhi (IIT Delhi)** *April 2025*
Research Intern Photonics New Delhi, India

- Designed and optimized Si_3N_4 waveguides for dual wavelength operation at 700 nm (visible) and 1550 nm (telecom).
- Integrated the optimized waveguide to realize a high Q ring resonator, evaluating through port and drop port transmission spectra using simulation sweeps.
- *Simulation Environment: Lumerical MODE, FDTD*

Awards & Achievements

- **Best Research Paper Award Symposium on Advanced Photonics (SAP 2025), IIT Indore**
  - Research: *Analyzing High Efficiency Hybrid Silicon HfO_2 ITO based Electro Optic Modulator*
  - Presented at SAP 2025, IIT Indore.
  - Developed a high performance, ring resonator based EOM featuring optimized ITO and HfO_2 layers.
  - Achieved modulation efficiency of 1.07 nm/V, extinction ratio 11.12 dB at 4 V.
  - Optimized: 6000 nm radius, 20 nm ITO, 20 nm HfO_2, 80 nm coupling gap.
  - Proved concept viability for low power, CMOS compatible operations.
  - *Simulation Tools: Lumerical MODE, FDTD, CHARGE, INTERCONNECT*

Personal Projects

- **Design & Simulation of Waveguide Grating Antenna for Optical Phased Arrays** *November 2025*
  - Designed a silicon on insulator (SOI) waveguide grating antenna for precise beam steering targeting LiDAR and free space optical communication at 1550 nm.
  - Wrote Python (lumapi) scripts to automate Ansys Lumerical FDTD and MODE workflows, reducing simulation setup time and enabling rapid parameter sweeps.
  - Optimized grating period and duty cycle through systematic parameter sweeps to achieve single mode TE operation with controlled effective index.
  - Analyzed far field radiation patterns and field decay for directionality verification, and built a custom 3D visualization tool to model beam steering.

Technical Skills and Interests

- **Languages**: Python, C++
- **Scientific Computing**: Python (NumPy, SciPy for numerical analysis), Matplotlib (Data Visualization).
- **Areas of Interest**: Optics, Photonics, Silicon Photonics, Electro Optic Modulation, Quantum Key Distribution
- **Simulation Tools**: Lumerical MODE, FDTD, CHARGE, INTERCONNECT, Zemax OpticStudio

Certifications

- **CENSE Winter School 2024**
- **NPTEL Certification in Photonic Integrated Circuits 2025**